Financial Transactions and Loans Dashboard

Using Microsoft Power BI

Project Report

Submitted By

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Tool Used: Microsoft Power BI Desktop

1. Introduction

The Financial Transactions and Loans Dashboard was created using Microsoft Power BI to present important banking information in a clear and interactive format. The dashboard brings together customer, account, loan and transaction information so that users can understand the overall position without checking each table separately.

The report includes key performance indicators (KPIs), account-type comparisons, transaction trends, loan-status distribution and a transaction summary. Slicers allow users to filter the report by year, account type and transaction type. Drill-down and tooltip features provide additional detail when exploring the visuals.

2. Objectives

To summarise customer, account, loan and transaction information in one report.

To display key figures through KPI cards.

To compare accounts across account types and transactions across transaction types.

To understand the distribution of loans by status.

To explore transaction activity over time using a date hierarchy.

To make the dashboard interactive with slicers, visual interactions and tooltips.

To practise data cleaning, data modelling and DAX measures in Power BI.

3. Dataset Description

The project uses Dataset 4, imported into Power BI as 11 tables. The tables cover customer details, customer types, addresses, accounts, account types, account statuses, loans, loan statuses, transaction types, transactions and branches.

Data area	Purpose in the dashboard
Customers and customer types	Identify and count customers and connect customers to their account records.
Accounts, account types and account statuses	Analyse account totals, balances, types and active accounts.
Loans and loan statuses	Analyse loan counts, principal amounts, interest rates and loan status.
Transactions and transaction types	Count transactions, total transaction value and transaction activity over time.
Addresses and branches	Provide address and branch information linked to the relevant records.

The main numeric fields used in calculations include account balance, loan principal amount and transaction amount. Date fields include account opening date, loan dates and transaction date.

4. Data Preparation

The data was imported into Power BI and cleaned in Power Query before creating the report. Table names were formatted consistently by capitalising the first letter.

4.1 Cleaning steps

Accounts: removed duplicate Account ID values, changed Opening Date to Date format, filtered out null opening dates, and changed Balance to fixed decimal number.

Addresses: removed duplicate Address ID values and replaced missing Street, City and Country values with suitable “unknown” labels.

Customers: removed duplicate CustomerID values and replaced missing first and last names with “unknown”. DOB was transformed and converted to Date using the English (India) locale; two invalid/missing values were replaced with null.

Loans: removed duplicate LoanID values, changed PrincipalAmount to fixed decimal number, and converted StartDate and EstimatedDate to Date.

Transactions: changed Amount to fixed decimal number, converted TransactionDate to Date, and applied Trim and Clean to Description.

Null handling was done carefully. Missing account opening dates were filtered because they represented incomplete or invalid account setup data. Missing dates of birth were

retained as nulls because removing those customer rows would also remove two customers who had active bank accounts.

4.2 Data relationships

Twelve relationships were created between the tables. Most relationships use one-to-many cardinality and single-direction filtering. The main active relationships connect customer types to customers, customers to accounts, account types and account statuses to accounts, loan statuses to loans, transaction types to transactions, branches to transactions, and accounts to transactions through AccountOriginID.

Some relationships were left inactive to avoid ambiguity and to allow special calculations when required: Addresses to Customers, Accounts to Loans, and Accounts to Transactions through AccountDestinationID. An automatically created one-to-one relationship between Branches and Addresses was deleted.

5. DAX Measures Used

DAX measures were created to calculate key values dynamically. The main measures were checked using a temporary table visual to see whether blank results appeared; the temporary table was then removed.

Measure	DAX formula
Total Customers	Total Customers = DISTINCTCOUNT(Customers[CustomerID])
Total Accounts	Total Accounts = DISTINCTCOUNT(Accounts[AccountID])
Total Account Balance	Total Account Balance = SUM(Accounts[Balance])
Active Accounts	Active Accounts = CALCULATE(DISTINCTCOUNT(Accounts[AccountID]), 'Account Statuses'[StatusName] = "Active")
Total Loans	Total Loans = DISTINCTCOUNT(Loans[LoanID])
Total Loan Principal	Total Loan Principal = SUM(Loans[PrincipalAmount])
Average Interest Rate	Average Interest Rate = AVERAGE(Loans[InterestRate])
Active Loans	Active Loans = CALCULATE(DISTINCTCOUNT(Loans[LoanID]), 'Loan Statuses'[StatusName] = "Active")

Total Transactions	Total Transactions = DISTINCTCOUNT(Transactions[TransactionID])
Total Transaction Value	Total Transaction Value = SUM(Transactions[Amount])
Destination Transaction Value	Destination Transaction Value = CALCULATE(SUM(Transactions[Amount]), USERELATIONSHIP(Accounts[AccountID], Transactions[AccountDestinationID]))
Total Loan Amount	Total Loan Amount = CALCULATE(SUM(Loans[Principal Amount]), USERELATIONSHIP(Accounts[AccountID], Loans[AccountID]))

The two additional measures use USERELATIONSHIP to calculate values through relationships that are inactive by default. This allows analysis through the destination account and the account connected to a loan when those calculations are needed.

6. Dashboard Components

6.1 KPI cards

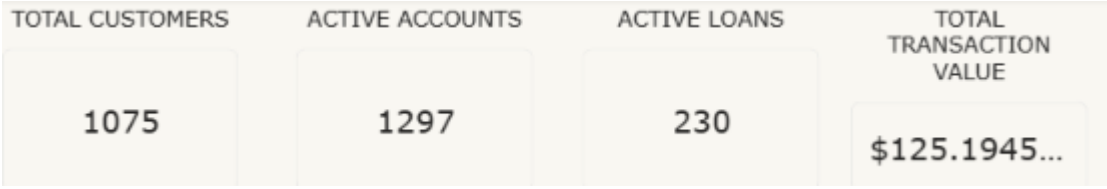
Four cards were placed at the top of the Finance Overview page to give a quick summary of the dataset.

Total Customers: 1,075

Active Accounts: 1,297

Active Loans: 230

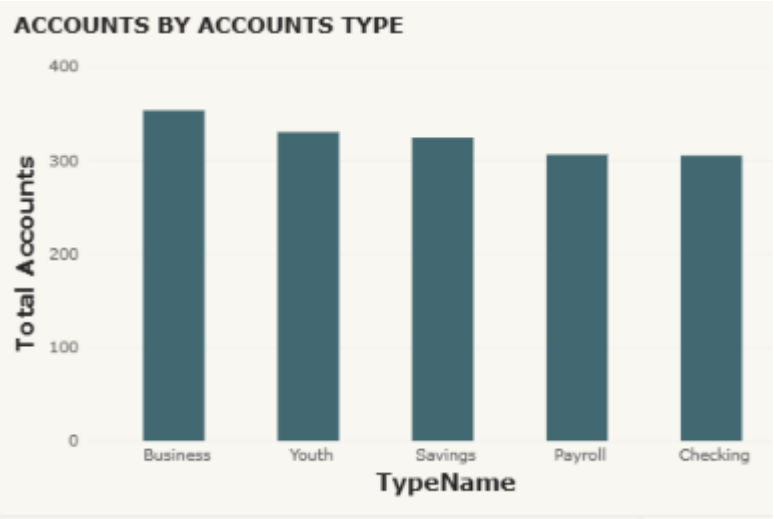
Total Transaction Value: approximately $125.954 million, as displayed on the dashboard.



6.2

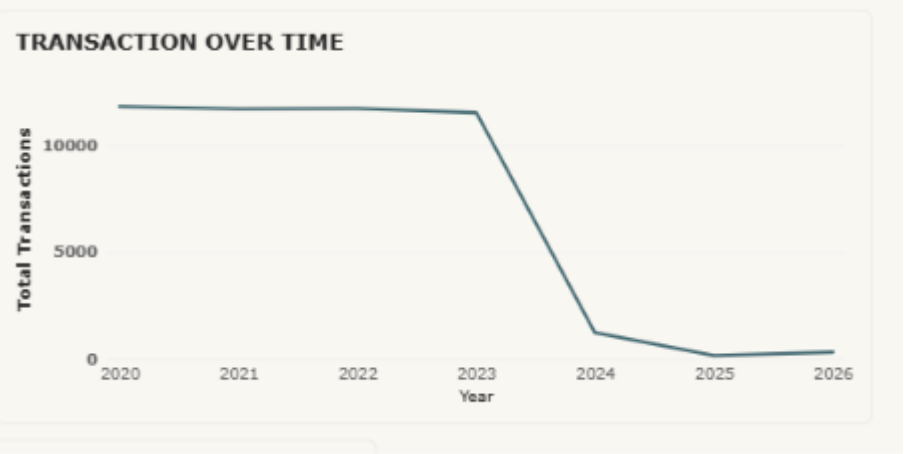
Accounts by Account Type – Clustered Column Chart

This chart compares the number of accounts across Business, Youth, Savings, Payroll and Checking account types. Business is the highest category in the displayed chart. The chart helps users compare the relative size of each account category.



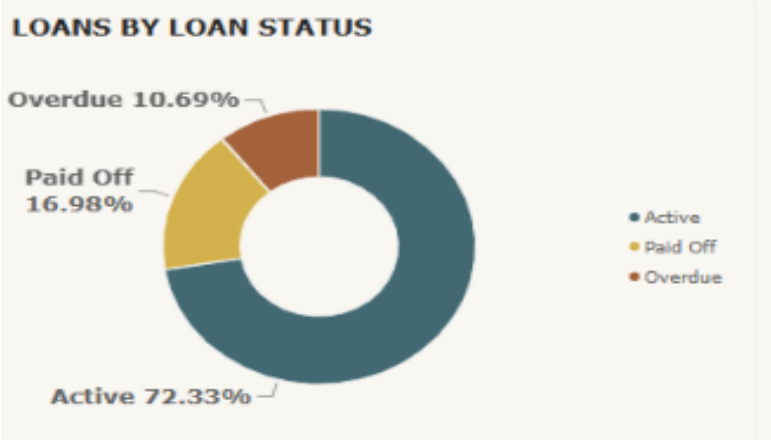
6.3 Transaction Over Time – Line Chart

The line chart displays transaction activity across time. The date hierarchy supports Year, Quarter and Month levels, allowing users to move from a broad yearly view to a more detailed period. The visual covers years shown from 2020 to 2026.



6.4 Loans by Loan Status – Doughnut Chart

The doughnut chart presents the distribution of loans by status. The dashboard shows Active loans at 72.33%, Paid Off loans at 16.98% and Overdue loans at 10.69%. Category labels and percentages are displayed to make the distribution easier to understand.



6.5 Transactions by Type – Clustered Bar Chart

The clustered bar chart compares the number of transactions by transaction type. The displayed counts are:

Deposit: 15,065 transactions

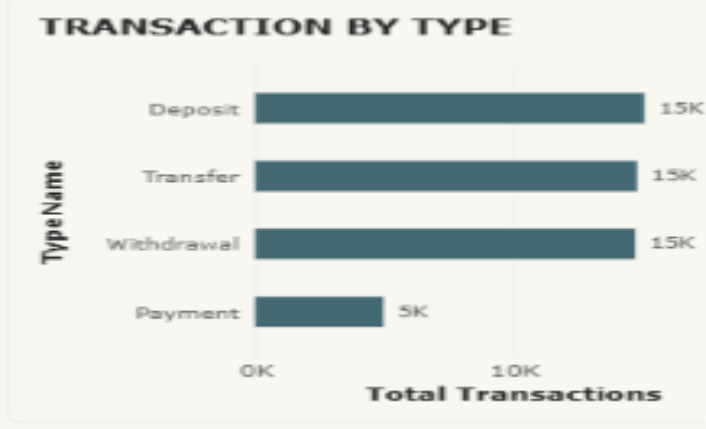
Transfer: 14,771 transactions

Withdrawal: 14,706 transactions

Payment: 4,958 transactions

Deposit has the highest count among the displayed types, while Payment has the lowest.

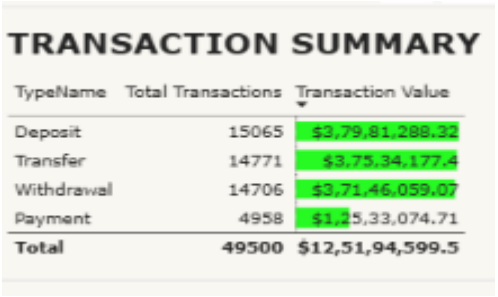
The displayed transaction counts add up to 49,500.



6.6 Transaction Summary – Table

The table contains Transaction Type Name, Total Transactions and Total Transaction Value. It gives users a more detailed view than the charts by showing the count and monetary value for each transaction type. Conditional formatting data bars were applied

to the transaction value column, with green for positive values and red for negative values as configured in the report.



TRANSACTION SUMMARY

TypeName	Total Transactions	Transaction Value
Deposit	15065	$3,79,81,288.32
Transfer	14771	$3,75,34,177.4
Withdrawal	14706	$3,71,46,059.07
Payment	4958	$1,25,33,074.71
Total	**49500**	**$12,51,94,599.5**

7. Slicers Used

Three slicers were added to the dashboard:

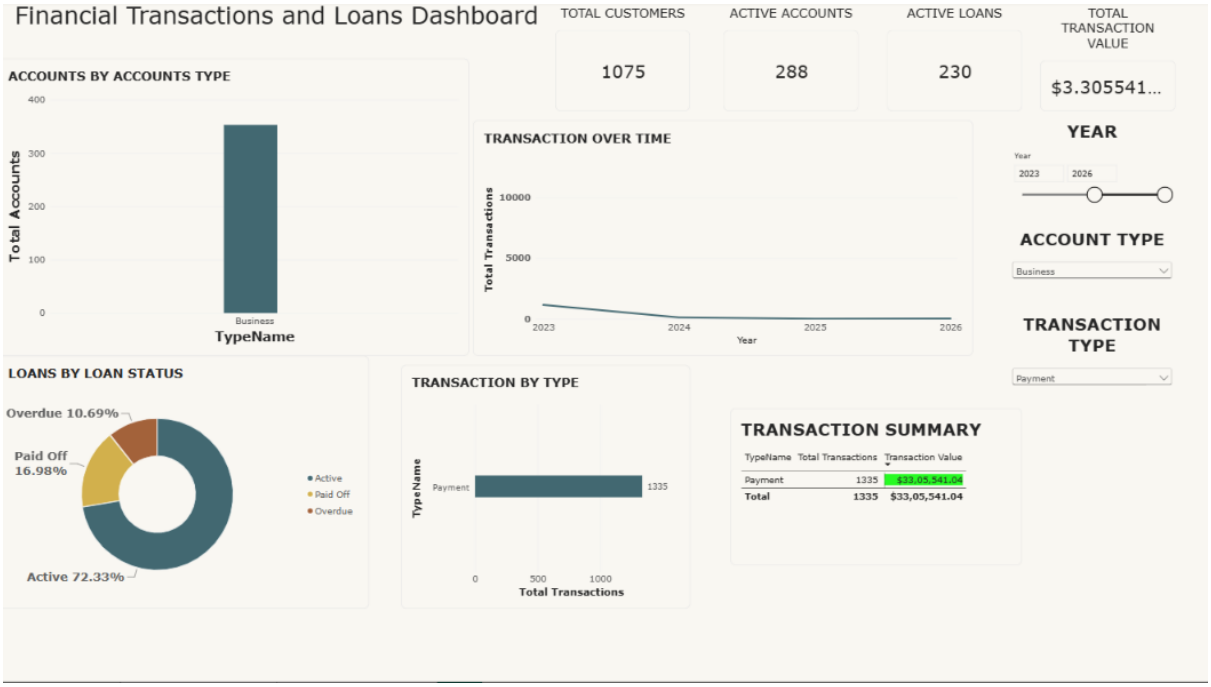
Year – filters the report to a selected year.

Account Type – filters by account category and was configured as a dropdown.

Transaction Type – filters by the selected transaction category and was configured as a dropdown.

The slicers make the report interactive. For example, selecting Business in the Account Type slicer changes the related account-type chart, transaction trend, transaction-type chart, transaction summary and KPI cards according to the selected filter context. Visual interactions were also reviewed using Edit Interactions, where filtering, highlighting or no interaction can be selected.





8. Key Findings

The dashboard reports 1,075 customers, 1,297 active accounts and 230 active loans.

The total transaction value displayed is approximately $125.954 million.

Business is the largest account category in the account-type chart.

Active loans form the largest share of the loan-status distribution at 72.33%. Paid Off loans account for 16.98%, while Overdue loans account for 10.69%.

Deposit is the most frequent displayed transaction type (15,065), followed by Transfer (14,771), Withdrawal (14,706) and Payment (4,958).

The transaction trend can be explored by year, quarter and month, supporting a more detailed view of transaction activity.

Slicers, tooltips and conditional formatting help users focus on selected categories and identify values more easily.

These findings describe the values visible in the dashboard. Further interpretation of causes or business performance would require additional context about the source data and business targets.

9. Conclusion

This project helped demonstrate how Microsoft Power BI can transform banking-related data into an interactive dashboard. The work involved importing and cleaning data,

creating relationships between tables, writing DAX measures and selecting suitable visuals for customers, accounts, loans and transactions.

The slicers allow users to explore selected years and categories, while drill-down reveals transaction activity at different time levels. The tooltip page provides extra information for transaction types, and conditional formatting makes transaction values easier to scan. Overall, the dashboard brings important information into one place and supports clearer review of transaction and loan data.

10. References

- Microsoft PowerBI Desktop
- Walmart Sales Dataset
- Microsoft PowerBI Documentation